

**CAMPINAS–MIAMI
MATHEMATICAL MEETING 2026**

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Conference information.

Campinas–Miami Mathematical Meeting 2026
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Organizing Committee.

Leonardo Cavenaghi, Lino Grama, Ludmil Katzarkov, Jaqueline Mesquita, Ricardo Miranda, Misha Verbitsky.

1. COLLAPSING FLAT $SU(2)$ -BUNDLES TO SPHERICAL 3-MANIFOLDS

Speaker.

Eder M. Correa (Campinas, Brazil)

Title.

Collapsing Flat $SU(2)$ -Bundles to Spherical 3-Manifolds.

Abstract.

In this talk, we present a geometric mechanism for the emergence of spherical 3-manifolds from the superspace of Riemannian metrics associated with flat $SU(2)$ -bundles over closed orientable hyperbolic surfaces. We show that any spherical 3-manifold can be realized as a boundary point in the Gromov-Hausdorff closure of this superspace through a collapsing process. Furthermore, we discuss how the convergence toward the spherical limit is controlled by the order of the fundamental group of the target manifold and the systolic geometry of the hyperbolic base. Finally, we illustrate these results by showing how the Bolza surface provides the optimal error bound for the convergence toward the Poincaré homology sphere.

Main theme.

Spherical 3-manifolds appear as Gromov–Hausdorff boundary points of superspaces of suitable $SU(2)$ -bundles over closed hyperbolic surfaces.

Spherical manifolds.

Definition 1.1. A Riemannian manifold M^n is *spherical* if it is complete and locally isometric to the round sphere S^n . Equivalently, it has constant sectional curvature $+1$ and universal cover S^n .

If M is connected, then

$$M \cong S^n/\Gamma$$

where $\Gamma < \text{Isom}(S^n)$ is finite and acts freely.

In dimension 3 we use

$$S^3 \cong \text{SU}(2),$$

via the unit quaternions. Thus one gets spherical quotients

$$\text{SU}(2)/\Gamma$$

when $\Gamma < \text{SU}(2)$ is finite and acts by left multiplication.

The finite subgroups of $\text{SU}(2)$ are the binary cyclic, binary dihedral, binary tetrahedral, binary octahedral and binary icosahedral groups. This is the ADE list through the McKay correspondence.

See the constant curvature discussion in Section 7, especially Equation 7.3.1. For the ADE side, see Section 3.

Warning.

The general spherical space-form statement is S^3/Γ , with $\Gamma < \text{SO}(4)$ finite and free. The ADE statement is the clean single-action case $\Gamma < \text{SU}(2)$.

Main theorem.

Theorem 1.2. *Let*

$$S = \text{SU}(2)/\Gamma$$

be a spherical 3-manifold, with $\Gamma < \text{SU}(2)$ finite. Then there exists a closed orientable hyperbolic surface Σ and a flat principal $\text{SU}(2)$ -bundle

$$\text{SU}(2) \rightarrow P \rightarrow \Sigma$$

such that S occurs as a boundary point of the Gromov–Hausdorff closure of

$$\mathcal{S}(P) = \text{Met}(P)/\text{Diff}(P).$$

Equivalently, there are smooth metrics g_i on P such that

$$(P, d_{g_i}) \rightarrow (S, d_{g_0})$$

in Gromov–Hausdorff distance.

Here g_0 is the spherical quotient metric. I am recording the theorem in the form understood from the talk. The exact statement should be checked against the speaker’s paper.

The point is that a fixed smooth manifold P can carry metrics which collapse, in the Gromov–Hausdorff sense, to a lower dimensional spherical space form.

Holonomy and monodromy.

Let $\pi : P \rightarrow M$ be a principal G -bundle with connection θ . Fix $m \in M$ and $u \in P_m$. If γ is a loop based at m , let $\tilde{\gamma}$ be its horizontal lift starting at u . Then $\tilde{\gamma}(1)$ lies in P_m , so there is a unique $g_\gamma \in G$ with

$$\tilde{\gamma}(1) = u \cdot g_\gamma.$$

The holonomy group is

$$\text{Hol}_u(\theta) = \{g_\gamma\}.$$

The restricted holonomy group $\text{Hol}_u^0(\theta)$ is obtained from null-homotopic loops. For a connected base it is the identity component of the holonomy group.

The monodromy map is

$$m_\theta : \pi_1(M, m) \rightarrow \text{Hol}_u(\theta) / \text{Hol}_u^0(\theta), \quad [\gamma] \mapsto [g_\gamma].$$

So the quotient by restricted holonomy records the discrete part of holonomy seen by the fundamental group.

Ambrose–Singer theorem.

Let Ω_θ be the curvature form of θ . The Ambrose–Singer theorem says that

$$\mathfrak{hol}_u(\theta) = \text{Lie } \text{Hol}_u^0(\theta)$$

is generated by curvature. More exactly, take a point $v \in P$ reachable from u by a horizontal curve, and take horizontal vectors $X, Y \in T_v P$. Then the curvature value

$$\Omega_\theta(v)(X, Y) \in \mathfrak{g}$$

transported back to u contributes to $\mathfrak{hol}_u(\theta)$. The span of all such values is the holonomy Lie algebra.

For a vector bundle connection ∇ , this is the formula

$$\mathfrak{hol}_x(\nabla) = \text{span}\{P_\gamma^{-1} R_y(X, Y) P_\gamma\}.$$

Here $\gamma : x \rightarrow y$, P_γ is parallel transport, and $X, Y \in T_y M$.

Thus curvature gives the infinitesimal part of holonomy, while monodromy records the residual discrete part.

Riemann–Hilbert correspondence.

For the smooth discussion here, the relevant case is the flat case:

$$\Omega_\theta = 0.$$

Then parallel transport depends only on the homotopy class of the path. Choosing $m \in M$ and $u \in P_m$ gives the holonomy representation

$$\rho_\theta : \pi_1(M, m) \rightarrow G, \quad [\gamma] \mapsto g_\gamma.$$

The Riemann–Hilbert correspondence identifies flat principal G -bundles with representations of the fundamental group:

$$\{\text{flat principal } G\text{-bundles on } M\} / \cong \longleftrightarrow \text{Hom}(\pi_1(M, m), G) / G.$$

The group G acts on representations by conjugation.

Thus two flat bundles with connection (P, θ) and (Q, ω) are equivalent if there exists a principal G -bundle isomorphism

$$\varphi : P \rightarrow Q$$

such that

$$\theta = \varphi^* \omega.$$

On the representation side this says that the holonomy representations ρ and ρ' are conjugate:

$$\rho' = g \rho g^{-1}.$$

If a point in the fiber over m is fixed, the conjugacy ambiguity disappears.

Conversely, a representation

$$\rho : \pi_1(M, m) \rightarrow G$$

defines a flat principal bundle by

$$\widetilde{M} \times_{\rho} G.$$

This is the flat case of the Riemann–Hilbert dictionary.

Collapse sequence.

The paper uses a Kaluza–Klein type family of metrics

$$g_{P,n} = \frac{1}{n} \pi^*(h_{\Sigma}) + (\theta, \theta)_{\mathrm{SU}(2)}$$

on the total space P . As $n \rightarrow \infty$, the horizontal directions shrink and one expects the Gromov–Hausdorff limit to be the spherical quotient determined by the holonomy.

Gromov–Hausdorff distance.

Let (X, d_X) and (Y, d_Y) be compact metric spaces. A correspondence is a subset

$$R \subset X \times Y$$

whose two coordinate projections are surjective.

A correspondence is like a generalized graph. The graph of a map $f : X \rightarrow Y$ is a correspondence only when f is surjective. In general, R behaves like a multivalued graph.

The distortion of R is

$$\mathrm{dis}(R) = \sup\{|d_X(x, x') - d_Y(y, y')| : (x, y), (x', y') \in R\}.$$

The Gromov–Hausdorff distance is

$$d_{GH}(X, Y) = \frac{1}{2} \inf_R \mathrm{dis}(R),$$

where R runs through all correspondences between X and Y .

A small-distortion correspondence is an almost distance-preserving matching.

Isometry classes.

The Gromov–Hausdorff distance is invariant under isometry. If

$$X \cong X', \quad Y \cong Y',$$

then

$$d_{GH}(X, Y) = d_{GH}(X', Y').$$

Hence d_{GH} descends to isometry classes:

$$d_{GH}([X], [Y]) = d_{GH}(X, Y).$$

Moreover, for compact metric spaces,

$$d_{GH}([X], [Y]) = 0 \iff X \cong Y.$$

So compact metric spaces modulo isometry form a genuine metric space.

Superspace.

Let P be a smooth manifold and denote by $\mathrm{Met}(P)$ the space of smooth Riemannian metrics on P . The diffeomorphism group acts by pullback. The superspace of P is

$$\mathcal{S}(P) = \mathrm{Met}(P) / \mathrm{Diff}(P).$$

Thus two metrics define the same point when they differ by a diffeomorphism.

Assume now that P is compact. Each metric g defines the compact metric space (P, d_g) , so there is a map

$$\mathcal{S}(P) \rightarrow \mathcal{GH}, \quad [g] \mapsto [(P, d_g)].$$

It is well-defined because if $g = \phi^* g'$, then

$$\phi : (P, d_g) \rightarrow (P, d_{g'})$$

is an isometry.

Conversely, by Myers–Steenrod, an isometry between smooth Riemannian manifolds is smooth. Thus superspace embeds in the Gromov–Hausdorff moduli space of compact metric spaces.

Riemannian submersion lemma.

A technical lemma used in the proof gives a Riemannian submersion

$$f : (P, g) \rightarrow (\mathrm{SU}(2)/\Gamma, g_0).$$

Here g_0 is the quotient of the round metric on $\mathrm{SU}(2) \cong S^3$.

Recall that $f : (P, g) \rightarrow (N, h)$ is a Riemannian submersion if f is a submersion and

$$df_p : (\mathrm{Ker} df_p)^\perp \rightarrow T_{f(p)}N$$

is an isometry for every $p \in P$. Thus

$$T_p P = \mathrm{Ker} df_p \oplus (\mathrm{Ker} df_p)^\perp$$

is the vertical-horizontal splitting.

For horizontal vectors X, Y one has

$$g(X, Y) = g_0(df(X), df(Y)).$$

This lemma is the bridge between the principal-bundle construction and the Gromov–Hausdorff collapse: the ADE quotient is not only a holonomy target, but also the metric quotient sitting below (P, g) .

Examples.

The Poincare homology sphere is

$$\Sigma(2, 3, 5) \cong S^3/I^*$$

where $I^* < \mathrm{SU}(2)$ is the binary icosahedral group of order 120. It double covers the usual icosahedral rotation group $A_5 < \mathrm{SO}(3)$. Thus the quotient is by the binary icosahedral group, not by the ordinary icosahedral group.

It is a homology sphere:

$$H_*(\Sigma(2, 3, 5), \mathbb{Z}) \cong H_*(S^3, \mathbb{Z}).$$

but it is not simply connected. Its fundamental group is I^* .

The Bolza surface is the smooth completion of

$$y^2 = x^5 - x.$$

It is a two-fold cover of $\mathbb{P}^1$ branched over

$$0, \infty, 1, -1, i, -i.$$

By Riemann–Hurwitz, a double cover of $\mathbb{P}^1$ branched over six points has genus

$$g = \frac{6-2}{2} = 2.$$

The Bolza surface is the unique *global* maximum of the systole function on the moduli space $\mathcal{M}_2$, after giving every Riemann surface its uniformizing hyperbolic metric. In other words, among closed hyperbolic surfaces of genus 2, it maximizes the length of the shortest closed geodesic.

Points to check.

- (1) Is the theorem stated for all spherical 3-manifolds, or only for the single-action quotients $SU(2)/\Gamma$?
- (2) In the construction, which representation $\rho : \pi_1(\Sigma) \rightarrow \Gamma$ is chosen?
- (3) What is the exact collapse sequence g_i on P ?
- (4) Where is the Riemannian submersion lemma used in the proof?

2. UPDATES ON FLOWS OF GEOMETRIC STRUCTURES

Speaker.

Henrique Sa Earp (Campinas, Brazil)

Title.

Updates on flows of geometric structures.

Abstract.

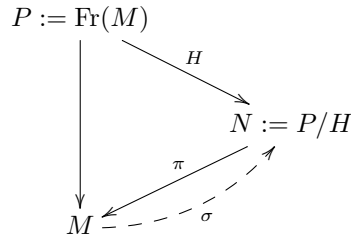
Aiming at a public with interests among Riemannian and complex geometry, Lie groups and parabolic PDEs, I will recall some results towards a general analytical theory for flows of H -structures, obtained with Fadel, Loubeau and Moreno. Then, as a concrete example, I will present some recent progress on $SU(2)$ -flows on 4-manifolds, initiating a classification programme of “parabolic” flows, based on a representation-theoretic method originally due to Bryant in the context of G_2 geometry, obtained with Fowdar.

Baby summary.

The talk asks which geometric structures can be moved by a well-posed PDE. First one sets up the general H -structure framework. Then one studies $SU(2)$ examples and uses representation theory to list the allowed flow terms.

H-structures.

Definition 2.1. Let M^n be a smooth manifold and let $H \subset GL(n, \mathbb{R})$. An H -structure on M is a reduction of the frame bundle $\text{Fr}(M)$ from $GL(n, \mathbb{R})$ to H . Equivalently, it is a section of $\text{Fr}(M)/H$. If ξ_0 is a reference tensor, then $H = \text{Stab}(\xi_0)$.



and there is an associated tensor bundle picture

$$\mathcal{F} \subset \bigoplus \mathcal{T}^{p,q} \rightarrow M, \quad \pi^* \mathcal{F} \rightarrow N.$$

Here TP splits into vertical and horizontal parts. Write

$$\begin{array}{ccc}
 TP & & \\
 \downarrow & \searrow & \\
 M & \xleftarrow{\pi} & TN := \nu \oplus \mathcal{H} \\
 & \nearrow \sigma &
 \end{array}$$

to record the tangent-bundle picture.

In prose: an H -structure is a way to equip M with extra geometric data that is modeled on a fixed tensor or algebraic structure with symmetry group H . The section $\sigma : M \rightarrow P/H$ picks, at each point, an adapted frame modulo H . Equivalently, one can encode the same information by a tensor field ξ on M whose stabilizer is exactly H ; the bundle P/H is the space of all such local choices.

Example 2.2. For an H -structure σ with torsion tensor T , one may consider the energy

$$E(\sigma) = \int_M |T|^2.$$

The harmonic flow is the negative gradient flow of this energy. Its purpose is to deform σ toward a more special or torsion-free structure.

Then it is natural to consider the Ricci flow of H -structures.

Definition 2.3. Let $H \subset \mathrm{SO}(n)$. A harmonic H -structure is a critical point of

$$E(\xi) = \frac{1}{2} \int_M |T|^2.$$

Here ξ is the corresponding H -structure and T is its torsion tensor.

Theorem 2.4 (ε -regularity). *Let (M, g) be closed and let $\xi(t)$ be a harmonic H -flow on $M \times [0, \tau)$. Fix $0 < r_M < \mathrm{inj}(M, g)$ and $E_0 \in (0, \infty)$. If the local energy of $\xi(t)$ on a geodesic ball $B_{r_M}(y)$ is small enough, then torsion stays controlled on a smaller ball for a short time.*

Theorem 2.5 (Energy gap). *Let (M, g) be a closed Riemannian n -manifold admitting a compatible H -structure. Then there is a constant $\varepsilon(M, g, H) > 0$ such that if a compatible harmonic H -structure ξ satisfies*

$$E(\xi) = \|\nabla \xi\|_{L^2(M)}^2 < \varepsilon(M, g, H),$$

then ξ is torsion-free:

$$\nabla \xi = 0.$$

The upshot is that the flow is being used to drive the structure into the small-energy regime where the ε -regularity theorem forces torsion to disappear. So the long-time picture is meant to end at a torsion-free limit.

One can also reverse the procedure. Let (M, g) be such that there is no torsion-free structure in the homotopy class. Then the question becomes whether the flow still has a useful limit, or whether it must break down.

I will recall results toward a general theory for flows of H -structures, obtained with Fadel, Loubeau and Moreno. Then I will present recent results on $\mathrm{SU}(2)$ -flows on 4-manifolds, starting a classification programme for parabolic flows. The

method is representation-theoretic and goes back to Bryant in G2 geometry; the recent work is joint with Fowdar.

The key concept to construct these flows is the Fadel–Loubeau–Moreno– SE 24 theorem, and also the Karigiannis 08 theorem: the “diamond action”. In this language the evolution equation is

$$\partial_t \eta = A \diamond \eta$$

with

$$A \equiv A(t) = S(t) + C(t), \quad S(t) \in S^2, \quad C(t) \in \Lambda^2.$$

The freedom in choosing A is the freedom to choose a differential invariant of the H -structure. Bryant’s 2003 point is that, for $H \subset O(n)$, the space $V_k(\mathfrak{h})$ of k th order invariants is characterized by

$$V_k(\mathfrak{h}) \oplus (\Lambda^1 \otimes S^{k+1}) = (S^2 \oplus \mathfrak{h}^\perp) \otimes S^k,$$

where $S^k = S^k(\Lambda^1)$ and $\mathfrak{h}^\perp = \Lambda^2/\mathfrak{h}$. The first space $V_1(\mathfrak{h}) = \Lambda^1 \otimes \mathfrak{h}^\perp$ is the intrinsic torsion. The next piece $V_2(\mathfrak{h})$ is spanned by ∇T and Rm , with overlap coming from the Bianchi-type identity

$$\nabla T \diamond \eta = \nabla^2 \eta + \text{l.o.t.}$$

so the lower-order terms are not independent.

In the simplest case $H = O(n)$, one has

$$V_1(\mathfrak{so}(n)) = 0.$$

Also,

$$V_2(\mathfrak{so}(n)) = \mathbf{R} \oplus S_0^2(\mathbf{R}^n) \oplus W,$$

and

$$\text{End}(\mathbf{R}^n)_{\mathfrak{so}(n)} = \mathbf{R} \oplus S_0^2(\mathbf{R}^n).$$

SU(2) flows.

Any $SU(2)$ -flow can be written as

$$\partial_t \omega = A \diamond \omega.$$

The point of the theory is that the possible choices of A form a finite dimensional space of differential invariants. So one can classify the flows. For $SU(2)$ this leads to a list of quadratic torsion functionals. There are 15 of them, built from terms of the form

$$\int_M g(a_i, a_j) \text{vol} \quad \text{and} \quad \int_M g(a_i, J_j a_k) \text{vol}.$$

Among them are the Dirichlet energy

$$\int_M |T|^2 \text{vol}$$

and the complex-geometric energy

$$\int_M |N_J|^2 \text{vol}.$$

So the classification problem is really a representation-theoretic bookkeeping problem: write down the allowed invariant tensors, then choose a linear combination that gives a parabolic flow.

In other words, the flow is not arbitrary. It is built from the finite list of quadratic expressions that the representation theory allows. The hard part is to choose the

coefficients so that the evolution equation is still geometric and parabolic. That is why the talk keeps returning to invariant tensors, torsion, and curvature.

Favorite flow.

Consider the two-parameter rich flow

$$\partial_t \omega = \left(-2 \operatorname{Ric}(g) - \lambda_1 \mathcal{L}_{\sum_{i=1}^2 J_i a_i} g + \lambda_2 \sum_{i=1}^3 \operatorname{div}(a_i) \right) \diamond \omega.$$

It has short-time existence and uniqueness when

$$-4\sqrt{5} < \lambda_2 + \lambda_1/2 < 0.$$

This is the favorite flow.

Questions and outlook.

Can one add terms involving W^+ to the flow and still keep short-time existence? Are there good choices of such curvature terms? The Lie-derivative term can be more generally taken as

$$\sum_{i,j} \lambda_{ij} \mathcal{L}_{J_i a_j} g.$$

What lower-order terms should one choose? Are there flows preserving torsion classes, for instance Laplacian-type flows? What about long-time existence, singularity formation, and solitons? More broadly, can the community classify the good flows for other groups H ?

Audience questions.

Is one tacitly assuming the flow is quasi-linear? Is that crucial?

The answer is that one wants to control low-order derivatives of the torsion. Once those are under control, the rest follows. So the point is to control the derivatives of the norm of the torsion.

Would it make sense to study flows that are not quasi-linear? For example, one could take the Riemann curvature tensor, contract it in many ways, and ask what happens.

Why would one do that? The lower-order terms depend on which PDE one chooses. The hope is to use this as technology for solving geometric PDEs. That is the perspective here.

Baby summary.

The talk is about moving geometric structures by PDEs. The structure has torsion, and the flow tries to reduce it. Representation theory tells you which terms you are allowed to use. In the best cases, the flow is well-posed and drives the structure toward something more rigid. The $SU(2)$ case is especially rich, because there are many explicit quadratic terms to choose from.

3. ATOM APPLICATIONS AND MMP

Speaker.

Ludmil Katzarkov

Title.

Atom applications and MMP.

Abstract.

In this talk we introduce the theory of Atoms. Applications and connection with MMP will be discussed.

Birational invariants.

- (1) LG-modules.
- (2) Orlov spectra.
- (3) Differential equations to the monads.
- (4) Road map to the twistor examples.

Orlov spectrum.

Take a generator g of $D^b(X)$ and generate by taking sums, summands, triangles and shifts. Let $t(g)$ be the minimum number of triangles needed to get to any object in $D^b(X)$.

The Orlov spectrum is the list of generation times of $D^b(X)$. In the shorthand from the talk,

$$D^b(X) = \langle t_1(g), \dots, t_n(g) \rangle.$$

- (1) $D^b(\mathbb{P}^1)$ has Orlov spectrum $(1, 2)$.
- (2) $D^b(E)$ has Orlov spectrum $(1, 2, 2, 4)$.

Conjecture: this number is the dimension of the variety. This was confirmed in dimension 1.

There are gaps in the spectrum. A conjecture by Ballard, Favero and Katzarkov says that if the gap is greater than $\dim X - 2$, then X is not rational.

Differential equations.

Now we take a shortcut and use differential equations. Keep in mind the Landau–Ginzburg model, but something else will happen. Take a Fano manifold X and consider it as an algebraic variety, but also as a symplectic manifold.

Main objective.

If X is a variety over $\mathbb{C}$, then birational geometry asks when

$$\mathbb{C}(X) \cong \mathbb{C}(x_1, \dots, x_n), \quad n = \dim X.$$

This is the rationality problem in its simplest form.

Mirror symmetry.

For smooth X , homological mirror symmetry says

$$D^b(X) \leftrightarrow \text{FS}(Y, W),$$

where Y is an open symplectic manifold and W is a function. The category on the right is built from Lagrangian submanifolds, and morphisms are given by Floer homology.

Example.

For $X = \mathbb{P}^2$ one has

$$D^b(\mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

The mirror is

$$Y = (\mathbb{C}^*)^2, \quad W = x + y + \frac{1}{xy}.$$

The observation made by them is that the singularity theory on the symplectic side recovers, my guess, the birational data on the algebraic side.

Blow-up.

Now turn to birational geometry. What happens when you blow up a point of $\mathbb{P}^2$?

$$D^b(\mathrm{Bl}_p \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2), E \rangle,$$

and on the symplectic side we get I_8 (singularity).

Cubic threefold.

Next example: $X_3 \subset \mathbb{P}^4$. The classical approach is by studying

$$H^{2,1}(X_3)/H_3(X_3, \mathbb{Z}),$$

that is, the intermediate Jacobian. This gives a moduli space called the Theta divisor

$$\Theta.$$

Example 3.1. From the HMS point of view, we have a symplectic manifold and a potential

$$W = \frac{(x + y + z)^3}{xyz} + z.$$

Some singular K3 surfaces appear. There is also a complicated thing at infinity.

Here one writes

$$D^b(X_3) = \langle \mathcal{A}, E_1, E_2 \rangle.$$

Theorem 3.2 ((-, Prizholkovsky, 2005)). *Let X be an equidimensional Fano with $\mathrm{Pic} X = \mathbb{Z}$, and suppose X is not $\mathbb{P}^3$. Consider the monodromy action on near-by fibers of the symplectic side. The monodromy is a 3×3 matrix. If μ is strictly unipotent, then X is rational.*

Differential equations.

Now we shall take a shortcut: we use differential equations. Keep in mind the Landau–Ginzburg model, but something else will happen. Take a Fano manifold X and consider it as an algebraic variety, but also as a symplectic manifold.

We have de Rham cohomology, and also a noncommutative Hodge package

$$H_{nc} = \bigoplus_{\lambda_i} H_{\lambda_i}.$$

The formula for E_{λ_i} in the notes is too unclear to reconstruct exactly, so I do not guess it here.

Theorem 3.3 (Kontsevich, Pantev, Yu). *The object*

$$\bigoplus_{\lambda_i} H_{\lambda_i}$$

is a birational invariant.

The quantum differential equation is

$$\left(\frac{\partial}{\partial u} - \frac{K}{u^2} + \frac{G}{u} \right) \psi(u) = 0.$$

where

$$\psi = u^\delta g(1, \dots).$$

Theorem 3.4. *Let $X \subset \mathbb{P}^N$ be a Fano manifold of dimension ≥ 3 . Then*

$$\delta_{\max} = \dim X - 1 \cdots$$

If MMP works, and we can ... then you might be able to conclude that this 7-dimensional cubic is Picard as well.

G-equivariant Hodge atoms.

Part 1.

Review of coarse Hodge atoms.

Let $X/\mathbb{C}$ be a smooth projective variety of dimension at least 2. For instance, one can think of a compact Kähler manifold. Let $G < \text{Aut}(X)$ be a finite subgroup. We call X a G -variety if G acts generically freely on X . The pair (X, G) is non- G -linearizable if there is no linear G -action on $\mathbb{P}^n$ and no G -equivariant birational map between X and $\mathbb{P}^n$.

The equivariant version of the coarse Hodge atoms keeps track of the G -action on the Hodge package. The natural coarse version is obtained by decomposing the atomic package into G -isotypical pieces.

Application.

These invariants give obstructions to G -linearizability. In particular, they produce many examples of non- G -linearizable G -actions on projective varieties.

4. RECENT PROGRESSES IN THE THEORY OF ATOMS

Speaker.

Leonardo Cavenaghi

Title.

Recent progresses in the theory of atoms.

Abstract.

In this talk we report on new results related to the theory of atoms obtained in collaboration with L. Katzarkov and M. Kontsevich. They will appear in the text titled "9.5 lectures in the theory of atoms", in the final stage of preparation.

Baby upshot.

The point is to package the Hodge-theoretic and quantum information into atoms. These atoms carry both the representation-theoretic and birational content.

Relation with Mumford–Tate groups.

This fits the notes in 6. There the Mumford–Tate group controls Hodge symmetries. The same Hodge package is also developed in the complex-geometry notes. Here the Hodge atoms are a more refined bookkeeping device: they split the Hodge package into pieces on which $\text{Hod}(\nabla)$ and its equivariant variants act.

G-equivariant atoms and quantum cohomology.

Theorem.

For any multi-degree hypersurface

$$X := X(1, 1, 1, 1) \subset (\mathbb{P}^1)^4,$$

for instance the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, the invariant $\mathbb{Z}/2$ -action by swapping the first and second, and the third and fourth factors, is not $\mathbb{Z}/2$ -linearizable. The same holds for

$$X \times \mathbb{P}^1,$$

where $\mathbb{Z}/2$ acts trivially on the $\mathbb{P}^1$ factor.

Coarse atoms.

For this we introduce a coarse version of the theory of G -equivariant atoms.

Notation.

Take your variety and its rational cohomology

$$V := H^*(X, \mathbb{Q}),$$

seen as a supervector space.

- Coordinates:

$$\mathbf{t} = \{t_a\}$$

formal dual super coordinates.

- Novikov ring: to track effective curve classes $\beta \in \overline{\text{NE}}(X, \mathbb{Z})$. The small Novikov ring is

$$\mathcal{N}_X.$$

It is the ring of power series.

- Coefficient ring: formal power series.

The Gromov–Witten potential allows us to define a new product in this ring that recovers the classical Poincaré cup product.

- Classical part: intersection theory.
- Quantum correction: curve counting, enumerative geometry.

Big quantum cohomology.

- Space:

$$V_{\text{big}} = V \otimes_{\mathbb{Q}} \mathcal{N}_X.$$

- Product: third-order differential operator.
- Deformation: full deformation in $\mathbf{t}$.

Small quantum cohomology.

- Space:

$$V \otimes \mathcal{N}_X.$$

- Product: [missing]
- Deformation: [missing]

Example.

For $X = \mathbb{P}^2$ with basis

$$\{\Delta_0, \Delta_1, \Delta_2\} = \{1, H, P\},$$

the Gromov–Witten potential is

$$\phi_X(\mathbf{t}, q) = \cdots$$

and it has a positive radius of convergence.

Big quantum product relations.

Questions.

- (1) Analytic germ.

- (2) Tangent space product.
- (3) Logarithmic tangent bundle.

General logarithmic tangent bundle.

Idea: to have a germ of a manifold such that the tangent space is the cohomology vector space and the quantum product is an associative product in this vector space.

For $X/\mathbb{C}$ smooth projective, let

$$V := H^*(X, \mathbb{Q}).$$

Let Φ_X be the Gromov–Witten potential. Then the third derivatives $\partial^3 \Phi_X$ yield a unital, commutative, associative product on the tangent bundle of a germ

$$M \subset H^*(X, \mathbb{C}).$$

This is the Frobenius-manifold picture behind quantum cohomology.

The problem is that convergence is most of the time hypothetical. The strategy is to introduce an auxiliary non-Archimedean field $\mathbb{T}$ of characteristic 0 to ensure convergence. The idea is to replace the germ M by a manifold B_X where we do have convergence for the Gromov–Witten invariants. More precisely, for $b \in B_X$,

$$T_b B_X \simeq H^*(X, \mathbb{T}, *_b).$$

Definition 4.1. The manifold B_X is called an A -model manifold.

Now we put an Euler vector field on B_X . If $*$ is a tensor field on B_X which is actually a Higgs field, then Eu scales the product. That is,

$$\mathcal{L}_{\text{Eu}}(*) = *.$$

Equivalently, we have an automorphism

$$k_b := \text{End}_{\text{Eu}}(*_b) \in \text{End}(H^*(X, \mathbb{T})).$$

That is the starting point of atom theory, from our perspective. There are other starting points.

Disjoint unions.

Thom–Sebastiani principle. If a variety is a disjoint union

$$X \sqcup Y,$$

its Gromov–Witten potential uncouples as a sum of the factors:

$$\Phi_{X \sqcup Y} = \Phi_X \oplus \Phi_Y.$$

This is important because we need to study blow-ups, the main operations in birational geometry.

Theorem 4.2 (Iritani, 2003). *Let $\tilde{X}$ be the blow-up of X along a smooth subvariety $Z \subset X$ of codimension $r \geq 2$. There exists a canonical analytic identification between open domains*

$$\psi : B_{\tilde{X}} \supset U \xrightarrow{\sim} V \subset B_X \times B_Z \times \cdots \times B_Z.$$

Moreover, one can choose the open domains so that the Euler vector field and the product are preserved:

$$\psi^*(*) = *, \quad \psi^*(\text{Eu}) = \text{Eu}.$$

Hodge action and fixed locus.

$\text{Hod}(\mathbb{T})$ acts on B_X preserving the quantum product $*$ and the Euler field Eu .

On coordinates:

- U_1 (Néron–Severi): trivial action, corresponding to algebraic classes.
- U_2, U_{odd} : scaling by Hodge weights.

Definition 4.3. The Hodge fixed locus

$$B_X^{\text{Hod}} \subset B_X$$

is the subvariety where $\text{Hod}(\mathbb{T})$ acts trivially. Geometrically, it is the locus where all non- (p, p) coordinates vanish.

$$\mathcal{HS} \simeq \text{Rep } \text{Hod}$$

for $\mathbb{Q}$ -polarizable pure Hodge structures.

We can decompose the tangent space to B_X using generalized eigenspaces:

$$T_b B_X = \bigoplus_{\lambda \in \text{Spec}(k_b)} V_\lambda.$$

The catch is that the group is pro-reductive. This means we can write, in fact,

$$T_b B_X = \bigoplus_{\lambda \in \text{Spec}(k_b)} \bigoplus_{\gamma} V_\lambda \cap V_\gamma.$$

The summands

$$\bigoplus_{\gamma} V_\lambda \cap V_\gamma$$

are called *coarse Hodge atoms*.

Each atom α corresponds to a $\overline{\mathbb{Q}}$ -linear representation

$$E^\alpha$$

of $\text{Hod}_{\overline{\mathbb{Q}}}$, independent of the point b .

Numerical invariants.

The Hochschild grading $(p - q)$ on $E^\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$ defines:

- (1) Rank of Hodge classes:

$$\rho_\alpha = \dim(V_\lambda \cap H_{p,q}).$$

- (2) Hodge polynomial. It is characterized by the coefficient:

$$P_\lambda(t) = \dim(V_\lambda \cap H^{p-q=n}(X)).$$

Theorem 4.4 (KKPY Non-rationality criterion). *Let X be an n -dimensional smooth complex projective variety. If there is a coarse Hodge atom in the Hodge atomic composition of X that does not appear in the Hodge atomic composition of any variety of dimension $\leq n - 2$, then X is not rational.*

Presence of G -actions.

The action on the variety defines an action on the cohomology. Now one does not look at $\text{Hod}(\mathbb{T})$ alone, but at

$$\text{Hod}(\mathbb{T}) \times G,$$

which acts on B_X . Restricting to

$$B_X^{G \times \text{Hod}},$$

a coarse G -atom α is the isomorphism class of a $G \times \text{Hod}(\mathbb{1})$ -representation on $\mathcal{B}_{\lambda,b}$.

G -invariant Hodge classes.

$$\rho_\alpha^G := \dim_{\mathbb{C}} \left(E^\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{C} \right)^{G \times \text{Hod}}.$$

Equivariant constraint.

Since the algebraic unit 1 is always $G \times \text{Hod}$ invariant, we have

$$\rho_\alpha^G \geq 1.$$

Obstructions to G -linearizable actions.

Atomic constraints in low dimensions (≤ 1).

Atoms originating from points (G/H) or curves ($G \times_H C$) satisfy:

- Rank bound:

$$\rho_\alpha^G \in \{1, 2\}.$$

- Rigidity: if ρ_α^G is constant, there is only one invariant Hodge class.

Example.

The hypersurface

$$X(1, 1, 1, 1) \subset (\mathbb{P}^1)^4.$$

Rationality: X is rational, the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ along an elliptic curve E .

The G -action is $G = \mathbb{Z}/2$, swapping factors $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$.

The atomic decomposition of X yields three primary coarse G -atoms:

λ	ρ	ρ^G	$P_\alpha(t)$
16	1	1	1
4	4	2	4
0	5	3	$t + 3 + t^{-1}$

Note that

$$\rho_{\text{total}}^G = 1 + 2 + 3 = 6,$$

tracking the G -invariant Hodge classes.

The atom of the elliptic curve must contribute. It contributes with:

- points $(1, 1)$ or $(2, 1)$.
- curve E : $(2, 2)$.

Target atom $\lambda = 0$:

$$(\rho, \rho^G) = (5, 3).$$

If the action was linearizable, its atoms must be constructible from atoms of $\mathbb{P}^3$ and G -invariant subvarieties of dimension ≤ 1 .

The contradiction: ...

Theorem 4.5 (Rationality criterion). *If X is a smooth complex projective variety of dimension $n \geq 2$, and the maximal generic spectral dimension on the Hodge locus B_X^{Hod} satisfies*

$$\mu(X) > n - 2,$$

then X is not rational.

Quantum differential equation.

Over the non-Archimedean field, recall that we have a manifold and then we put the quantum differential equation. We solve it via an Ansatz. After a big analysis, we get for the cubic, indeed,

$$\frac{5}{3}.$$